

1. Write a computer program to identify the root for $y = 10x^5 - 9x^3 + 7x$ using newton-raphson method.
2. Write a computer program to identify the root for $y = 10x^5 - 9x^3 + 7x$ using the false-position method.
3. 3. Write a computer program to identify the root for $y = 10x^5 - 9x^3 + 7x$ using secant method.
- 4.

θ	10	15	20	25
y	52	25	12	10

For the above tabulated function find the value of $\theta = 17$ using forward and backward difference dividend formula. Are both methods ending in the same value? Why or not? Justify your answer.

Step 1: Corrected Difference Table

θ	y	Δy	$\Delta^2 y$	$\Delta^3 y$
10	52			
15	25	-27		
20	12	-13	14	
25	10	-2	11	-3

Calculation of u

For $\theta = 17$:

$$u = \frac{\theta - \theta_0}{h} = \frac{17 - 10}{5} = \frac{7}{5} = 1.4$$

Step 2: Forward Difference Formula

$$y(17) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0$$

Substituting values:

$$y(17) = 52 + 1.4(-27) + \frac{1.4(0.4)}{2}(14) + \frac{1.4(0.4)(-0.6)}{6}(-3)$$

Calculating:

$$= 52 - 37.8 + 3.92 + 0.168 = 18.288$$

Step 3: Backward Difference Formula

For $u = \frac{17-25}{5} = -1.6$:

$$y(17) = y_n + u\nabla y_n + \frac{u(u+1)}{2!}\nabla^2 y_n + \frac{u(u+1)(u+2)}{3!}\nabla^3 y_n$$

Substituting values:

$$y(17) = 10 + (-1.6)(-2) + \frac{(-1.6)(-0.6)}{2}(11) + \frac{(-1.6)(-0.6)(-0.4)}{6}(-3)$$

Calculating:

1. First Term:

$$= 10 + 3.2$$

2. Second Term:

$$= \frac{(-1.6)(-0.6)}{2}(11) = \frac{0.96}{2}(11) = 0.48 \cdot 11 = 5.28$$

3. Third Term:

$$= \frac{(-1.6)(-0.6)(-0.4)}{6}(-3) = \frac{0.384}{6}(-3) = 0.064 \cdot (-3) = -0.192$$

Combining:

$$y(17) = 10 + 3.2 + 5.28 - 0.192$$

$$= 10 + 8.288$$

$$= 18.288$$

Final Results

- Forward Difference: $y(17) \approx 18.288$
- Backward Difference: $y(17) \approx 18.288$
- Average Result: $y(17) \approx \frac{18.288+18.288}{2} \approx 18.288$

This summary captures the corrected calculations with 11 included properly. If you have further questions, feel free to ask!

5. Derive the Trapezoidal rule of integration and the related error. From the error analysis between the Trapezoidal rule, the Simpsons $\frac{1}{3}$, and $\frac{3}{8}$ rule, which one can be concluded as better?

Ans:

Derivation of the Trapezoidal Rule

The Trapezoidal Rule is used for approximating the definite integral of a function. Given a continuous function $f(x)$ over the interval $[a, b]$, we can approximate the area under the curve by dividing it into n equally spaced subintervals.

1. Divide the Interval:

- Let $h = \frac{b-a}{n}$ be the width of each subinterval.
- The points are $x_0 = a, x_1 = a + h, \dots, x_n = b$.

2. Approximate the Area:

- The area under the curve can be approximated as:

$$\int_a^b f(x) dx \approx \frac{h}{2} \left(f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right)$$

This formula uses the trapezoids formed at each subinterval, where the first and last terms are not multiplied by 2.

Error Analysis of the Trapezoidal Rule

The error in the Trapezoidal Rule can be expressed as:

$$E_T = \int_a^b f(x) dx - T_n$$

where T_n is the result from the Trapezoidal Rule. The error can be estimated by:

$$E_T = -\frac{(b-a)^3}{12n^2} f''(\xi)$$

for some $\xi \in (a, b)$. This indicates that the error decreases with the square of the number of subintervals n .

Comparison with Simpson's 1/3 and 3/8 Rules

1. Simpson's 1/3 Rule:

- For an even number of segments, it approximates the integral as:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left(f(x_0) + 4 \sum_{i=1, \text{odd}}^{n-1} f(x_i) + 2 \sum_{i=2, \text{even}}^{n-2} f(x_i) + f(x_n) \right)$$

- The error for Simpson's 1/3 rule is given by:

$$E_S = -\frac{(b-a)^5}{90n^4} f^{(4)}(\xi)$$

2. Simpson's 3/8 Rule:

- It uses a similar approach but approximates the curve with cubic polynomials:

$$\int_a^b f(x) dx \approx \frac{3h}{8} \left(f(x_0) + 3 \sum_{1, 2, 3} f(x_i) + 2 \sum_{4, 5, 6} f(x_i) + f(x_n) \right)$$

- The error is:

$$E_{3/8} = -\frac{(b-a)^5}{80n^4} f^{(4)}(\xi)$$

Conclusion on Error Analysis

- **Order of Errors:**

- The Trapezoidal Rule has an error of order $O\left(\frac{1}{n^2}\right)$.
- Both Simpson's 1/3 and 3/8 Rules have errors of order $O\left(\frac{1}{n^4}\right)$.

Conclusion

- **Which Rule is Better?**

- Given that both Simpson's rules have a higher order of accuracy (smaller error) than the Trapezoidal Rule, they are generally preferred for numerical integration when evaluating smooth functions.
- Among Simpson's 1/3 and 3/8, the choice depends on whether the number of intervals is even or a multiple of three, but both offer better accuracy than the Trapezoidal Rule.

In summary, for smoother functions, Simpson's rules (1/3 and 3/8) provide better approximations than the Trapezoidal Rule due to their lower error terms.